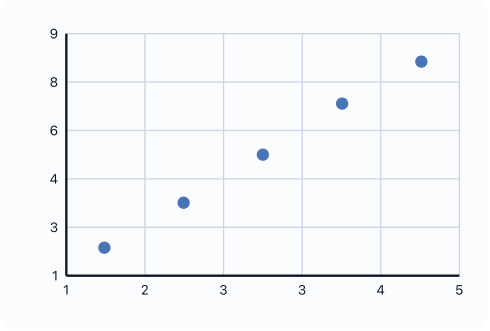
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Logistic Regression & Classification Practice Exercises

1 MCQ

In evaluating a binary classifier, the confusion matrix shows $TP = 45$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.



A 81.0%
B 81.8%
C 52.4%
D 13

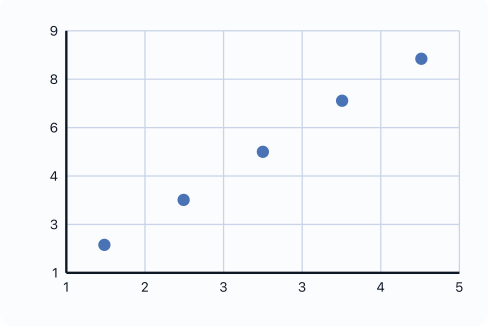
2 MCQ

In evaluating a binary classifier, the confusion matrix shows $TP = 55$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

A 82.6%
B 84.6%
C 56.5%
D 13

3 MCQ

In evaluating a binary classifier, the confusion matrix shows $TP = 60$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.



A 83.3%
B 85.7%
C 58.3%
D 13

4 MCQ

In evaluating a binary classifier, the confusion matrix shows $TP = 70$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

A 84.6%

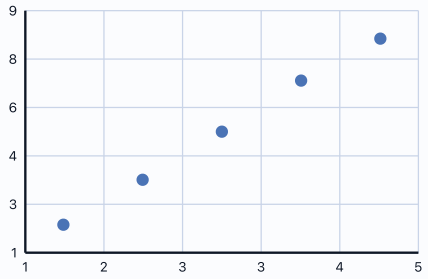
B 87.5%

C 61.5%

D 13

5 MCQ

A spam classification model identifies $TP = 30$ genuine spam emails and $FP = 10$ legitimate emails falsely flagged as spam. What is the precision of this model?



A 75.0%

B 60.0%

C 25.0%

D 85.7%

6 MCQ

A spam classification model identifies $TP = 60$ genuine spam emails and $FP = 20$ legitimate emails falsely flagged as spam. What is the precision of this model?

A 75.0%

B 60.0%

C 25.0%

D 85.7%

7 MCQ

A spam classification model identifies $TP = 90$ genuine spam emails and $FP = 30$ legitimate emails falsely flagged as spam. What is the precision of this model?

A 75.0%

B 60.0%

C 25.0%

D 85.7%

8 MCQ

A spam classification model identifies $TP = 120$ genuine spam emails and $FP = 40$ legitimate emails falsely flagged as spam. What is the precision of this model?

A 75.0%

B 60.0%

C 25.0%

D 85.7%

9 MCQ

Consider a simple logistic regression model where the decision boundary is defined by the linear equation $w_1x_1 + w_2x_2 + b = 0$. If the weights are $w_1 = 2$ and $w_2 = 3$, and the bias is $b = -7$, what is the equation of the decision boundary?

A $3x_1 + 2x_2 + 7 = 0$

B $2x_1 + 3x_2 + 7 = 0$

C $2x_1 + 3x_2 - 7 = 0$

D $-2x_1 - 3x_2 + 7 = 0$

10 MCQ

Consider a simple logistic regression model where the decision boundary is defined by the equation $-1.5x_1 + 2x_2 + 5 = 0$. For a data point (x_1, x_2), if $-1.5x_1 + 2x_2 + 5 < 0$, it is classified into Class 0. Otherwise, it is classified into Class 1. Which of the following points lies on the decision boundary?

A $(-1, 1)$

B $(2, -2)$

C $(0, -2.5)$

D $(3, 0.5)$

11 MCQ

Considering evaluating a binary classifier, the confusion matrix shows $TP = 55$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

Binary Classification Contingency Table

	Pred Positive	Pred Negative
Actual Positive	55	10
Actual Negative	10	40

A 56.5%

B 13

C 84.6%

D 82.6%

12 MCQ

Examining evaluating a binary classifier, the confusion matrix shows $TP = 60$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

Binary Classification Contingency Table

	Pred Positive	Pred Negative
Actual Positive	60	10
Actual Negative	10	40

A 85.7%

B 83.3%

C 58.3%

D 13

13 MCQ

For the configuration in evaluating a binary classifier, the confusion matrix shows $TP = 70$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

Binary Classification Contingency Table

	Pred Positive	Pred Negative
Actual Positive	70	10
Actual Negative	10	40

A 87.5%

B 84.6%

C 61.5%

D 13

14 MCQ

Consider a binary classification problem where the decision boundary is defined by the logistic function $p(x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2)}}$, which represents the probability of a data point $\mathbf{x} = (x_1, x_2)$ belonging to class 1. The estimated model parameters are $\beta_0 = -3$, $\beta_1 = 1$, and $\beta_2 = -2$. We are interested in the geometric shape of the classification boundary. The classification boundary occurs where $p(x) = 0.5$. Find the equation of this boundary in the x_1x_2-plane.

A $x_2 = \frac{1}{2}x_1 - \frac{3}{2}$

B $x_2 = 2x_1 - 3$

C $x_2 = 2x_1 + 3$

D $x_2 = \frac{1}{2}x_1 + \frac{3}{2}$

15 MCQ

Consider a simple logistic regression model where the decision boundary is defined by the equation $3x + 4y - 12 = 0$. This equation represents a linear boundary in a 2D feature space. What is the y-intercept of this decision boundary?

A $y = -4$

B $y = -3$

C $y = 4$

D $y = 3$

16 MCQ

A binary classification model uses a logistic sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$. The decision boundary is defined by $\sigma(z) = 0.5$, where $z = w_1 x_1 + w_0$. For a dataset with two features x_1 and x_2, the model's linear component is $z = 2x_1 - x_2 + b$. If the classification boundary passes through the point $(x_1, x_2) = (3, 1)$, what is the value of the bias b?

A $b = 7$

B $b = -5$

C $b = -7$

D $b = 5$

17 MCQ

In an applied problem, a spam classification model identifies $TP = 90$ genuine spam emails and $FP = 30$ legitimate emails falsely flagged as spam. What is the precision of this model?

A 25.0%

B 85.7%

C 75.0%

D 60.0%

18 MCQ

Suppose a spam classification model identifies $TP = 120$ genuine spam emails and $FP = 40$ legitimate emails falsely flagged as spam. What is the precision of this model?

A 85.7%

B 60.0%

C 75.0%

D 25.0%

19 MCQ

An observed spam classification model identifies $TP = 150$ genuine spam emails and $FP = 50$ legitimate emails falsely flagged as spam. What is the precision of this model?

A 60.0%

B 25.0%

C 75.0%

D 85.7%

20 MCQ

Consider a spam classification model identifies $TP = 180$ genuine spam emails and $FP = 60$ legitimate emails falsely flagged as spam. What is the precision of this model?

A 25.0%

B 75.0%

C 85.7%

D 60.0%

21 MCQ

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.30$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

A -0.847

B 0.429

C 0.353

D -2.347

22 MCQ

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.40$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

A 0.667

B 0.795

C -1.905

D -0.405

23 MCQ

In an applied problem, a logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.50$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

A -1.500

B 1.200

C 0.000

D 1.000

24 MCQ

Suppose a logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.60$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

A 0.405

B -1.095

C 1.605

D 1.500

25 MCQ

An observed logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.70$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

A -0.653

B 2.047

C 0.847

D 2.333